

ECON 0100 | Exercise B2 | Supply

| Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{Q_s}{10}$$

Prices are in galleons and quantity is in pasties.

Plot. Vertical intercept, $Q_s = 0$: $P = 2$. Below 2 galleons quantity supplied is zero, so the curve starts there and rises by 1 galleon for every 10 pasties. This is a flatter supply curve than the Vignette's, $P = 2 + \frac{1}{2}Q_s$, so the numbers below are bigger.

Q1 | Quantity Supplied

a) What is quantity supplied at 10 galleons? _____

80 pasties.

Price to quantity: plug the price into the supply curve and solve for Q_s .

$$10 = 2 + \frac{1}{10}Q_s \rightarrow 8 = \frac{1}{10}Q_s \rightarrow Q_s = 80$$

b) What is marginal cost at 9 pasties? _____

2.9 galleons per pasty.

Quantity to price: the height of the supply curve at 9 pasties is the marginal cost of the ninth pasty, the least a seller will take for it: $P = 2 + \frac{1}{10} \cdot 9 = 2.9$.

Q2 | Producer Surplus

What is producer surplus at 10 galleons? _____

PS = 320 galleons.

Producer surplus is the triangle above supply and below the price, from $Q = 0$ out to the 80 pasties sold in Q1. Its height is the price minus the 2-galleon intercept and its base is the quantity:

$$PS = h \cdot b \cdot \frac{1}{2} = (10 - 2) \cdot 80 \cdot \frac{1}{2} = 320$$

Revenue at 10 galleons is the rectangle $10 \cdot 80 = 800$ the 320 above the curve is surplus and the 480 below it covers marginal costs.

